Measure Equivalence as a Guide for Learning SPDE Priors from Replicated Fields

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Abstract

Learning a Gaussian prior from replicated spatial fields requires deciding not only how much flexibility to add to a reference model, but where it belongs: as a correction to the reference covariance, or as a change to the differential operator that defines it. Gaussian-measure equivalence theory separates these cases in the continuum, but on any fixed discretization all nondegenerate Gaussians are equivalent, so the distinction can only be seen under mesh refinement. We study this in a leaky-aquifer SPDE where uncertain pumping, spatially varying leakage, and heterogeneous conductivity enter through the source, a lower-order term, and the principal part of the operator. Only the last changes the measure class. From 200 replicated fields whitened against a fitted reference, a diagonal refinement statistic decays to zero for source and leakage departures but stays at the plateau (≈ 0.061) predicted for a conductivity lens. Models of the wrong class fail in characteristic ways. A stationary model absorbs source variability by driving its range to infinity ($\widehat{\kappa} \approx 3 \times 10^{-5}$ against a true $\kappa_0 = 6$), and correction-only models fitting a conductivity change incur a KL cost that grows linearly with resolution, so reference-based complexity measures become discretization dependent. A combined model that adapts the principal operator and learns a finite-rank source correction recovers $\widehat{\kappa} = 6.08$ on the source-only population and separates the conductivity field from the source subspace when both are present. Replicated data can therefore decide whether to correct a Gaussian reference or change its operator; deciding wrongly yields a well-fitting but structurally incorrect model.

1 Introduction

Gaussian random fields are widely used as probabilistic models for spatially varying quantities, and stochastic partial differential equation (SPDE) representations connect covariance structure to local differential operators (Lindgren et al., 2011, 2022). Replicated realizations carry information about population-level covariance beyond that in a single realization, and have been used to estimate nonstationary dependence and to separate persistent population structure from realization-specific variation (Ingebrigtsen et al., 2015; Fuglstad et al., 2015). We ask whether replicated fields can reveal if a covariance discrepancy can be represented as a correction of a chosen reference model, or whether the operator defining that reference must itself change.

A common response to mismatch is to add flexibility around a reference model. One class of methods augments a simpler covariance model with low-dimensional or low-rank structure, as in fixed-rank kriging, predictive processes, and full-scale covariance approximations (Cressie and Johannesson, 2008; Banerjee et al., 2008; Sang and Huang, 2012); LatticeKrig, nearest-neighbor Gaussian processes (NNGP), and fused Gaussian processes likewise operate on the correction side of a fixed reference. A second class modifies the covariance-generating operator, for example through spatially varying range, anisotropy, or coefficients in an

SPDE representation (Fuglstad et al., 2015; Lindgren et al., 2022). Penalized-complexity (PC) priors shrink a flexible component toward a simpler base model (Simpson et al., 2017; Fuglstad et al., 2019). In each case the reference is taken as given; whether it is the right reference for the observed population is not asked.

In function space these departures are not alike: by the Feldman–Hájek dichotomy and the operator results of Bolin and Kirchner (2023), smooth source and lower-order perturbations can remain in the Gaussian measure class of the reference, whereas a change of the principal coefficient cannot (Section 2). On a finite mesh, however, all of them appear only as covariance changes.

The distinction matters for methods that learn probability measures independently of discretization. Lim et al. (2025) formulate score-based diffusion in function space under absolute-continuity and Cameron–Martin conditions relative to a Gaussian perturbing measure, and Pidstrigach et al. (2023) show that discretize-first diffusion constructions can deteriorate under mesh refinement and develop an infinite-dimensional formulation with dimension-independent guarantees. In linear inverse problems the Gaussian posterior differs from its prior only through a finite-dimensional update, and Carere and Lie (2026) characterize low-rank posterior approximations that remain equivalent to both. Population-prior learning has no analogous guarantee that the observed population lies in the measure class of the chosen reference.

The statistical problem also differs from comparing two specified continuum models: the reference and a finite collection of replicates are observed, but the mechanism producing the discrepancy is not, and on a fixed discretization every pair of nonsingular Gaussians is mutually absolutely continuous. Fixed-domain results add a second difficulty: covariance parameters can be weakly identified while prediction remains accurate, and misspecified range parameters can produce asymptotically similar interpolation (Stein, 1999; Zhang, 2004; Kaufman and Shaby, 2013). The range parameter thus provides a weakly constrained low-frequency direction through which a stationary fit can absorb localized source variability, so a finite-resolution model can fit well while assigning variability to the wrong structural component.

Mechanism misattribution is already recognized in spatial statistics. Fuglstad et al. (2015) show that inadequately modeled mean structure can be absorbed by the covariance model, creating spurious long-range dependence or apparent nonstationarity, and replicated fields help separate competing sources of spatial variation (Ingebrigtsen et al., 2015). Cui and He (2027) derive identifiability conditions for the drift and diffusion of dynamical SPDEs from trajectory data; Bolin (2026) places known covariates in the forcing of an SPDE to separate pointwise and operator-mediated mean effects; and Petersson and Schroers (2025) estimate and test the driving-noise covariance with the differential operator treated as given. Here the question is instead whether a covariance discrepancy across replicated equilibrium fields belongs to the source covariance or to the operator, and how the alternatives behave under mesh refinement.

The same question precedes reference-based prior construction. PC priors formalize complexity as departure from a base model (Simpson et al., 2017), parameterized for Matérn fields so that the distance from the base remains meaningful (Fuglstad et al., 2019); for anisotropic SPDEs, principal-operator changes produce mutually singular measures and require alternatives to the Kullback–Leibler construction (Llamazares-Elias et al., 2026). Forcing a principal-part change into a reference-based correction still gives a finite KL discrepancy on a fixed mesh, but one that grows with resolution, so the implied complexity is itself discretization dependent.

We study the question in a leaky-aquifer SPDE, where uncertain pumping, spatially varying leakage, and heterogeneous conductivity enter through the source, a lower-order term, and the principal part of the operator. The object of study is the relative covariance T_N of the population with respect to the reference in reference-whitened coordinates (Section 2.2), estimated from replicates by its sample counterpart $\widehat{T}_N$. Its spectral and refinement behavior distinguishes source-like covariance corrections from persistent principal-part discrepancy.

The resulting modeling principle is to adapt the reference operator when the population exhibits principal-part variation and to represent finite-dimensional source variability as a correction relative to the adapted reference. We implement it in a combined model that learns a spatially varying principal coefficient together

with a low-rank covariance correction, each component selected separately from the data.

The contributions are fourfold. First, we use Gaussian-measure equivalence results to formulate a prior-learning problem in which source, lower-order, and principal-part departures are distinguished by their behavior under refinement, and show that a smooth principal-part departure is multiplication plus a compact operator, which fixes its refinement plateau and KL slope (Sections 2 and 5.1). Second, we develop a sample-based whitening and refinement diagnostic for persistent principal-part mismatch that remains informative when the departure produces no isolated eigenvalue outside the sample covariance bulk (Section 5.2). Third, we characterize the asymmetric misattribution failures produced when source variability is represented through the operator or principal-part variability through a correction-only model, including a KL cost that grows with resolution (Section 5.3). Fourth, we introduce a combined model that learns the principal operator and finite-dimensional source corrections jointly (Section 5.4).

2 Problem formulation and theoretical basis

This section defines the population-level distinction that is later inferred from finite replicated data: what separates a covariance correction that stays in the Gaussian measure class of a reference from a departure that changes the covariance-generating operator.

2.1 Reference field and population-learning problem

Let $\mathcal{D} \subset \mathbb{R}^2$ be a bounded domain and consider the elliptic SPDE

$$A_0 u = \mathcal{W}, \qquad A_0 = \kappa_0^2 I - \nabla\cdot(K_0\nabla)\,, \tag{1}$$

with homogeneous Dirichlet boundary conditions. Here $\mathcal{W}$ denotes spatial Gaussian white noise, κ_0^2 is a zeroth-order coefficient, and K_0 is the principal coefficient. The nominal reference used below has $K_0 \equiv 1$. At the continuum level, we take $\mathcal{W}$ to have identity covariance in the formal white-noise coordinates. Under the present boundary conditions A_0 is positive and self-adjoint, so the centered reference field has law

$$\mu_0 = \mathcal{N}(0, \Sigma_0), \qquad \Sigma_0 = A_0^{-1}A_0^{-1} = A_0^{-2}. \tag{2}$$

Equation (2) is a continuum operator identity. Its discrete form depends on the representation of white noise. For the finite-difference scheme used in the numerical study, nodal white noise has covariance $Q_W = h^{-2}I_N$, giving $\Sigma_{0,N} = \sigma^2 h^{-2} A_{0,N}^{-1} A_{0,N}^{-T}$. The complete discretization is specified in Section 4.

The value $\kappa_0 = 0$ remains admissible on a bounded domain with homogeneous Dirichlet conditions. In that case $A_0 = -\Delta_D$, whose smallest eigenvalue is strictly positive, so the inverse and the covariance remain well defined. This observation will matter when fitting stationary reference models, because decreasing κ corresponds to increasing the spatial range without making the bounded-domain covariance singular at $\kappa = 0$.

We assume that replicated fields

$$u^{(1)}, \ldots, u^{(n)} \overset{\text{iid}}{\sim} \mu_1 = \mathcal{N}(0, \Sigma_1) \tag{3}$$

are generated from an unknown population law. The prior-learning problem is to determine how Σ_1 differs structurally from the known reference covariance Σ_0. In particular, the population departure may arise through the stochastic source, through a lower-order term in the operator, or through the principal coefficient itself.

2.2 Relative covariance in reference-whitened coordinates

The comparison becomes transparent after removing the covariance already represented by the reference. Define

$$T = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2} - I. \tag{4}$$

If

$$z = \Sigma_0^{-1/2} u, \tag{5}$$

then the reference field satisfies $\mathrm{Cov}_{\mu_0}(z) = I$, while under the population law

$$\mathrm{Cov}_{\mu_1}(z) = I + T. \tag{6}$$

Thus T is the covariance structure that remains after whitening by the reference model.

On a discretization with N degrees of freedom, the corresponding matrix is

$$T_N = \Sigma_{0,N}^{-1/2} \Sigma_{1,N} \Sigma_{0,N}^{-1/2} - I_N. \tag{7}$$

Its eigenvalues $t_{j,N}$ quantify relative covariance departures mode by mode. The accumulated squared relative departure is

$$D_N := \|T_N\|_{\mathrm{HS}}^2 = \sum_{j=1}^{N} t_{j,N}^2. \tag{8}$$

The refinement behavior of D_N will provide the bridge between continuum equivalence and the finite-dimensional statistics used later.

2.3 Gaussian equivalence and the Hilbert–Schmidt condition

For two Gaussian measures on an infinite-dimensional Hilbert space, the Feldman–Hájek theorem gives an equivalence-or-singularity dichotomy. In one standard formulation, equivalence requires: (i) equality of the Cameron–Martin spaces, (ii) the mean difference to lie in that common space, and (iii) an appropriate relative covariance operator to be Hilbert–Schmidt (Da Prato and Zabczyk, 2014, Thm. 2.25). All fields considered here are centered, so the mean condition is automatic. For the elliptic families below, the common-domain and Cameron–Martin conditions are supplied by the corresponding operator results; the condition that distinguishes the cases of interest is therefore the Hilbert–Schmidt behavior of the relative covariance.

Under these conditions,

$$T \in \mathcal{S}_2 \qquad \Longleftrightarrow \qquad \|T\|_{\mathrm{HS}}^2 = \sum_{j=1}^{\infty} t_j^2 < \infty, \tag{9}$$

where $\mathcal{S}_2$ denotes the Hilbert–Schmidt class. We use this condition to distinguish departures whose cumulative relative effect decays sufficiently rapidly across fine scales from departures that persist through the high-frequency spectrum.

2.4 Three forms of population departure

The leaky-aquifer example provides three physically distinct perturbations of (1). For each we give the operator, its measure class, and the condition that decides it.

Source variability. Random pumping through r smooth profiles gives

$$A_0 u = \mathcal{W} + G\xi, \qquad G\xi = \sum_{k=1}^{r} \xi_k g_k, \qquad \xi \sim \mathcal{N}(0, C), \tag{10}$$

so $\Sigma_1 = A_0^{-1}(I + GCG^*)A_0^{-1}$ and, since $\Sigma_0^{1/2} = A_0^{-1}$, $T = GCG^*$ has rank at most r and is Hilbert–Schmidt. The Cameron–Martin space of $\mu_0 = \mathcal{N}(0, A_0^{-2})$ is $\mathrm{Ran}(A_0^{-1})$, so the induced field direction $A_0^{-1} g_k$ belongs to it exactly when $g_k \in L^2(\mathcal{D})$. Smooth well profiles satisfy this condition and idealized delta sources do not, so the smooth-source assumption is load-bearing for the finite-rank in-class construction.

Spatially varying leakage. Changing only the zeroth-order coefficient, $A_1 = -\nabla\cdot(K_0\nabla) + \kappa^2(x)$, remains in the reference class in the smooth setting of Bolin and Kirchner (2023, Thm. 4.11 and Table 1); T is Hilbert–Schmidt but generally not finite rank. With $V = \kappa^2(x) - \kappa_0^2$, to first order

$$T \approx -\left(VA_0^{-1} + A_0^{-1}V\right), \tag{11}$$

and for bounded smooth V the singular values inherit the elliptic decay $s_j(A_0^{-1}V) = O(j^{-2/d})$, so $\sum_j s_j^2 \sim \sum_j j^{-4/d}$ is finite for $d < 4$: infinitely many modes are affected, but with a relative effect that decays fast enough to remain Hilbert–Schmidt in $d \le 3$.

Spatially varying conductivity. Changing the principal coefficient, $A_1 = \kappa_0^2 I - \nabla\cdot(K_1(x)\nabla)$, leaves the class: in the variable-coefficient setting of Bolin and Kirchner (2023, Thm. 4.11 and Lemma 4.9), equivalence requires $K_1(x) = K_0(x)$. With the driving-noise normalization held fixed, even a constant rescaling $K_1 = cK_0$, $c \neq 1$, is excluded, since $(\kappa^2 - c\Delta)^{-2} = c^{-2}(\kappa^2/c - \Delta)^{-2}$ changes both covariance amplitude and range.

Physical departure	SPDE location	Relative covariance	Model component
Random smooth pumping	Source	Finite rank; in class	Low-rank correction
Spatially varying leakage	Zeroth order	HS; generally infinite rank ($d \le 3$)	Lower-order operator
Spatially varying conductivity	Principal part	Outside reference class	Principal operator

The variable-coefficient theorem assumes smooth coefficients and a smooth boundary. The main conductivity lens is smooth, but the unit square has corners, so we test whether the predicted refinement signature persists on the polygonal domain; a discontinuous disc lens is considered separately as a robustness experiment.

2.5 Refinement behavior of the relative covariance

The Hilbert–Schmidt condition gives an immediate finite-resolution consequence for in-class departures.

Proposition 2.1 (Bounded refinement energy for an in-class departure)**.** *Let $\{P_N\}_{N\ge1}$ be nested finite-rank orthogonal projections converging strongly to the identity, let $T \in \mathcal{S}_2$, and define*

$$T_N = P_N T P_N, \qquad D_N = \|T_N\|_{\mathrm{HS}}^2. \tag{12}$$

Then

$$D_N \longrightarrow \|T\|_{\mathrm{HS}}^2 < \infty, \qquad \frac{D_N}{N} \longrightarrow 0. \tag{13}$$

The proof is given in Appendix A. Proposition 2.1 formalizes the refinement signature of an in-class departure: its accumulated squared relative covariance approaches a finite limit.

For a principal-coefficient change, Bolin and Kirchner (2023, Lemma 4.9) establish the non-compactness that underlies failure of equivalence. Non-compactness implies that the relative perturbation does not decay in the manner required of a Hilbert–Schmidt operator, but it does not by itself imply a linear growth rate for D_N. The stronger scaling

$$D_N = O(N), \qquad \frac{D_N}{N} \longrightarrow c > 0, \tag{14}$$

follows from a persistent-mode mechanism: if there exist $\rho, \delta > 0$ such that at least ρN eigenvalues satisfy $|t_{j,N}| \geq \delta$, then

$$D_N = \sum_{j=1}^{N} t_{j,N}^2 \geq \rho N \delta^2. \tag{15}$$

For a smooth principal-coefficient change, the following lemma shows that this mechanism is present and identifies the limits.

Lemma 2.2 (A principal-part departure is multiplication plus compact)**.** *Let $\mathcal{D} \subset \mathbb{R}^2$ be bounded with either a smooth boundary or a convex polygonal boundary, let $K \in C^1(\overline{\mathcal{D}})$ with $K \geq k_0 > 0$, and let*

$$A_0 = \kappa^2 I - \Delta, \qquad A_1 = \kappa^2 I - \nabla \cdot (K \nabla), \tag{16}$$

with the same homogeneous Dirichlet conditions and $\Sigma_j = c\, A_j^{-2}$ for a common constant $c > 0$. Then

$$T = \Sigma_0^{-1/2} \Sigma_1 \Sigma_0^{-1/2} - I = M_m + R, \qquad m(x) = K(x)^{-2} - 1, \tag{17}$$

where M_m is multiplication by m and R is compact on $L^2(\mathcal{D})$. Consequently:

(i) $\sigma_{\mathrm{ess}}(T) = m(\overline{\mathcal{D}})$. *In particular T is not compact unless $K \equiv 1$.*

(ii) Let P_N project onto piecewise-constant functions on a partition of $\mathcal{D}$ into N cells of equal measure whose diameters tend to zero, and let $T_N = P_N T P_N$. Then

$$\frac{D_N}{N} \longrightarrow \frac{1}{|\mathcal{D}|} \int_{\mathcal{D}} m^2 \, dx, \qquad \frac{1}{N} \sum_{i=1}^{N} (T_N)_{ii}^2 \longrightarrow \frac{1}{|\mathcal{D}|} \int_{\mathcal{D}} m^2 \, dx, \tag{18}$$

so the off-diagonal energy per node vanishes.

(iii) The empirical eigenvalue distribution of T_N converges weakly to the distribution of $m(X)$ with X uniform on $\mathcal{D}$. Hence the persistent-mode condition holds with $\rho = |\{|m| \geq \delta\}|/|\mathcal{D}|$, and

$$\frac{\mathrm{KL}_N}{N} \longrightarrow \frac{1}{2|\mathcal{D}|} \int_{\mathcal{D}} \big[m - \log(1+m) \big] \, dx. \tag{19}$$

The proof is given in Appendix B. Part (i) recovers the non-compactness of Bolin and Kirchner (2023, Lemma 4.9); parts (ii) and (iii) give the extensive scaling (14) with an explicit constant, which equivalence theory alone does not provide. The lemma is stated for a piecewise-constant Galerkin compression; the finite-difference relative covariance used in the numerical study is a consistent approximation of it, and the agreement of the two is checked numerically in Section 5.1. The proof requires $K \in C^1$ and does not cover the discontinuous disc lens of Section 5.5; there the per-node energy also persists ($D_N/N \approx 0.035$), which suggests that the conclusion extends, but this is not proved here.

The same distinction appears in Gaussian relative entropy. If the eigenvalues of $I + T_N$ remain bounded away from zero, then

$$\mathrm{KL}_N(\mu_1 \,\|\, \mu_0) = \frac{1}{2} \sum_{j=1}^{N} \left[t_{j,N} - \log(1 + t_{j,N}) \right]. \tag{20}$$

For a bounded Hilbert–Schmidt perturbation this sum can converge, whereas a persistent $O(1)$ shift over a positive fraction of modes produces an $O(N)$ KL cost; for a smooth principal-coefficient change, Lemma 2.2(iii) gives its slope. Thus a principal-part mismatch can appear finite at every fixed discretization while becoming increasingly distant from the reference as the mesh is refined.

2.6 What the continuum theory does not determine

These results compare known population covariances along a refinement sequence, whereas in prior learning $\Sigma_{1,N}$ is unknown and only the sample covariance $\widehat{\Sigma}_{1,N} = n^{-1} \sum_i u^{(i)} u^{(i)\top}$ of n centered replicates is available, giving $\widehat{T}_N = \Sigma_{0,N}^{-1/2} \widehat{\Sigma}_{1,N} \Sigma_{0,N}^{-1/2} - I_N$. When N/n is not small, sampling variability dominates: even under the exact reference, the sample covariance S of n draws from $\mathcal{N}(0, I_N)$ satisfies $\mathbb{E}\|S - I_N\|_F^2 = N(N+1)/n$. The naive sample Hilbert–Schmidt norm therefore grows with dimension under the null and cannot serve directly as a refinement diagnostic. The continuum theory identifies *what* must be distinguished but not *how* to distinguish it from a finite ensemble, which is the subject of Section 3.

3 Statistical methodology

Section 2 identifies the distinction to be inferred: bounded refinement energy for a Hilbert–Schmidt departure, against non-vanishing energy per resolved degree of freedom for a persistent principal-part departure. We first diagnose the structure of the departure from replicated fields, which must contend with finite-ensemble fluctuations, and then fit covariance models whose components can represent that structure.

3.1 Diagnosing covariance departure from finite replicates

3.1.1 Estimated reference and whitening

Let

$$\widehat{\Sigma}_{0,N} = \Sigma_{0,N}(\widehat{\sigma}^2, \widehat{\kappa}) \tag{21}$$

denote the stationary reference covariance obtained by maximum likelihood from the replicated fields. We use this fitted stationary model as a *reference for whitening*, not as the final covariance model. If the discrete reference covariance is written in the general form

$$\Sigma_{0,N} = \sigma^2 A_\kappa^{-1} Q_W A_\kappa^{-T}, \tag{22}$$

with Q_W the discrete white-noise covariance, a corresponding whitening map is

$$z^{(j)} = \widehat{\sigma}^{-1} Q_W^{-1/2} A_{\widehat{\kappa}} u^{(j)}, \qquad j = 1, \ldots, n. \tag{23}$$

For the finite-difference discretization used in the numerical study, $Q_W = h^{-2} I$ and (23) reduces to $z = h A_{\widehat{\kappa}} u / \widehat{\sigma}$.

The whitened sample covariance is

$$S = \frac{1}{n} \sum_{j=1}^{n} z^{(j)} z^{(j)\top}. \tag{24}$$

If the reference parameters were known, $\mathbb{E}[S] = I + T_N$. Because the same replicated fields are used to estimate $(\widehat{\sigma}^2, \widehat{\kappa})$, the resulting diagnostic is a plug-in statistic rather than an exactly unbiased estimate of the relative covariance. This distinction is relevant when the stationary range parameter is weakly identified or moves toward the boundary. Such range misspecification remains an in-class covariance deformation, however, and therefore cannot by itself generate a nonzero per-node refinement plateau of the type sought below. We assess the finite-sample sensitivity of the diagnostic to reference fitting separately in the numerical study.

3.1.2 Spectral outliers and finite-dimensional source structure

A finite-rank source perturbation produces isolated population eigenvalues in the whitened covariance,

$$I + T = I + \sum_{k=1}^{r} c_k v_k v_k^\top, \qquad c_k > 0. \tag{25}$$

At finite N and n, even the null covariance produces a nontrivial sample spectrum. Let

$$\gamma = \frac{N}{n}. \tag{26}$$

For whitened Gaussian noise, the upper Marchenko–Pastur edge (Marchenko and Pastur, 1967) is

$$\lambda_+ = \left(1 + \sqrt{\gamma}\right)^2. \tag{27}$$

We therefore report the spectral-outlier count

$$\widehat{r}_{\mathrm{MP}} = \#\left\{\widehat{s}_k > \lambda_+\right\}, \tag{28}$$

where $\widehat{s}_k$ are the nonzero eigenvalues of S in decreasing order. The count is used as a diagnostic for isolated positive covariance directions; the fitted correction rank is selected independently by cross-validation. A spike of strength c_k separates from the bulk only above the BBP threshold $c_k > \sqrt{\gamma}$ (Baik and Silverstein, 2006).

The outlier count alone cannot diagnose a principal-part departure. A spatially distributed operator change can alter many population modes without producing an isolated sample eigenvalue outside the Marchenko–Pastur bulk. This motivates a second statistic that accumulates a local signature over the spatial discretization.

3.1.3 Diagonal refinement statistic h_N

Let

$$d_i = \frac{1}{n} \sum_{j=1}^{n} z_{ij}^2 \tag{29}$$

denote the empirical variance of the whitened field at node i. For a fixed whitening reference,

$$\mathbb{E}[d_i] = 1 + T_{ii}. \tag{30}$$

Under Gaussian sampling,

$$d_i \overset{d}{=} (1 + T_{ii}) \frac{\chi_n^2}{n}, \tag{31}$$

and therefore

$$\mathbb{E}\left[(d_i - 1)^2\right] = T_{ii}^2 + \frac{2}{n}(1 + T_{ii})^2, \tag{32}$$

$$\mathbb{E}[d_i^2] = \frac{n+2}{n}(1 + T_{ii})^2. \tag{33}$$

This gives the bias-corrected statistic

$$\widehat{H}_N = \sum_{i=1}^{N} \left[(d_i - 1)^2 - \frac{2}{n+2} d_i^2\right], \tag{34}$$

for which, conditional on a fixed whitening reference,

$$\mathbb{E}[\widehat{H}_N] = \sum_{i=1}^{N} T_{ii}^2. \tag{35}$$

We use the normalized quantity

$$\widehat{h}_N = \frac{\widehat{H}_N}{N}. \tag{36}$$

The statistic is motivated by the distinction in Section 2. If T is Hilbert–Schmidt, then for any orthonormal basis

$$\sum_i T_{ii}^2 \le \|T\|_{\mathrm{HS}}^2 < \infty, \tag{37}$$

so the corresponding per-node diagonal energy vanishes under refinement. A principal-coefficient change has a different local high-frequency signature. By Lemma 2.2, the relative covariance of a smooth scalar conductivity perturbation is multiplication by

$$m(x) = K(x)^{-2} - 1 \tag{38}$$

plus a compact remainder. Consequently, under regular refinement (Lemma 2.2(ii)),

$$\frac{1}{N}\sum_{i=1}^{N} T_{ii}^2 \longrightarrow \frac{1}{|\mathcal{D}|}\int_{\mathcal{D}} \left[K(x)^{-2} - 1\right]^2 \, dx =: h_\infty, \tag{39}$$

for the piecewise-constant compression of the lemma; for the finite-difference grid basis used here the limit is a prediction checked against the oracle values. For the conductivity lens specified in Section 4, $h_\infty = 0.060732$.

Equation (35) is exact only when the whitening reference is fixed. Estimating the stationary reference from the same replicates introduces plug-in error, and estimating and removing source directions from the same sample introduces an additional dependence. We therefore refer to (34) as a bias-corrected diagnostic rather than an exactly unbiased test after fitting.

3.1.4 Deflating isolated source directions

Mixed populations can contain both finite-dimensional source variability and a distributed principal-part departure. To expose the latter, we optionally remove isolated positive spectral directions before evaluating (34). Let

$$\widehat{V}_r = [\widehat{v}_1, \ldots, \widehat{v}_r] \tag{40}$$

contain the sample eigenvectors associated with eigenvalues above the upper Marchenko–Pastur edge. The whitened fields are projected onto the orthogonal complement,

$$Z_\perp = (I - \widehat{V}_r \widehat{V}_r^\top) Z. \tag{41}$$

Projection reduces the marginal variance at node i by the leverage

$$\ell_i = \sum_{k=1}^{r} \widehat{v}_{ik}^2. \tag{42}$$

The deflated nodal variance is therefore renormalized as

$$d_i^\perp = \frac{n^{-1} \sum_j z_{ij,\perp}^2}{1 - \ell_i}, \tag{43}$$

and substituted into (34).

In the experiments considered here, random sources add positive covariance directions while the conductivity lens is *stiff*, $K(x) \geq 1$, so its leading relative-covariance contribution is non-positive. Under this sign structure, upper-edge outliers provide a clean way to identify source directions before evaluating the principal-part diagnostic. For a soft lens, $K(x) < 1$, a principal-part change can itself create positive high-frequency variance; the upper-edge count is then not mechanism-specific, although the diagonal refinement statistic h_N remains sign-insensitive. The present deflation rule should therefore be interpreted as a source-removal step for the sign structure studied here, not as a universal classifier of all principal-part perturbations.

Because $\widehat{V}_r$ is estimated from the same sample used to compute $d_i^\perp$, the deflated statistic is only approximately bias corrected. A split-sample or cross-fitted implementation can remove this dependence and is used as a robustness check when evaluating the diagnostic.

3.2 Covariance model classes

We compare model families that place flexibility at different levels of the SPDE. All are special cases of

$$\Sigma = \sigma^2 A^{-1} Q_W^{1/2} \left(I + VCV^\top\right) Q_W^{1/2} A^{-T}, \qquad V^\top V = I, \quad C = \mathrm{diag}(c_1, \ldots, c_r),\ c_k \geq 0, \tag{44}$$

and differ in the operator A and in whether the whitened finite-rank correction $VCV^\top$ is present (Table 3.1). M1 is the stationary reference $A_\kappa = \kappa^2 I + L$ with no correction; it defines the fitted reference used for whitening and the baseline for all comparisons. M2 keeps the eigenvectors of L and reshapes the stationary modal variances by $\exp[-\delta(\lambda_j)]$, with δ a cubic B-spline expansion in $\log \lambda$, as a flexible stationary control. M3 adds the correction to the stationary base; for a fixed base, the maximum-likelihood directions are the leading eigenvectors of the whitened sample covariance, with strengths from the corresponding excess variances. M4-κ replaces κ^2 by the spatial field $\kappa^2 \exp[\Phi\theta]$, and M4-$K$ replaces L by L_{K_θ} with $\log K_\theta = \Phi\theta$, where the columns of Φ are smooth radial basis functions; M4-K changes the reference operator itself and is the class able to represent principal-part structure.

The combined model M5 uses both components:

$$\boxed{\Sigma_{\mathrm{M5}} = \sigma^2 A_\theta^{-1} Q_W^{1/2} \left(I + VCV^\top\right) Q_W^{1/2} A_\theta^{-T}, \qquad A_\theta = \kappa^2 I + L_{K_\theta}.} \tag{45}$$

The operator A_θ represents population variability that changes the principal spatial mechanism, while $VCV^\top$ represents finite-dimensional source variability relative to the adapted operator. M5 therefore implements the rule suggested by Section 2: change the reference operator where the population requires a principal-part change, and represent isolated source variability as a correction around that reference.

Table 3.1: Model classes, parameterizations and fitting protocols. Rank is selected by five-fold held-out log likelihood where applicable; numerical tolerances are listed in Table C.1.

Model	Intended departure	Representation	Dimension / rank	Fitting	Initialization / regularization
M1	stationary reference (baseline)	$\sigma^2 h^{-2}(\kappa^2+\lambda)^{-2}$ in the DST basis	2 parameters	L-BFGS-B; σ^2, κ joint	$\sigma^2 = 1$, $\kappa = 6$; no penalty
M2	stationary mis-specification (control)	M1 $\times e^{-\delta(\lambda)}$, with six cubic B-splines in $\log\lambda$	8 parameters	L-BFGS-B; joint fit	M1 fit, $\theta = 0$; no penalty
M3-joint	random sources (correction)	M1 base plus PSD spikes in whitened coordinates	$2 + r(N+1)$; $r = 0, \ldots, 8$ by CV	Alternating PCA and L-BFGS-B; whitened noise fixed at 1 in PCA block	Two starts: M1-based and $(1, \kappa_0)$; no penalty
M4-κ	leakage (lower-order operator)	$\kappa(x)^2 = \kappa^2 \exp(\Phi\theta)$; 25 Gaussian RBFs	27 parameters	L-BFGS-B with JAX gradients; σ^2 profiled	$\kappa = \hat{\kappa}_{\mathrm{M3}}$, $\theta = 0$; $\lambda_\theta \lVert\theta\rVert^2$
M4-K	conductivity (principal operator)	$\log K = \Phi\theta$; same 25 RBFs	27 parameters	L-BFGS-B with JAX gradients; σ^2 profiled	$\kappa = \hat{\kappa}_{\mathrm{M3}}$, $\theta = 0$; $\lambda_\theta \lVert\theta\rVert^2$
M5	mixed source/operator (combined)	M4-K base plus PSD whitened spikes	$27 + r(N+1)$; $r = 0, \ldots, 5$ by CV	Exact block coordinate ascent: PPCA spike block and L-BFGS operator block	Flat K, $\kappa = \hat{\kappa}_{\mathrm{M3}}$; $\lambda_\theta \lVert\theta\rVert^2$

For M3-joint, the M1-based start uses $(\hat{\sigma}^2_{\mathrm{M1}}, \max(\hat{\kappa}_{\mathrm{M1}}, 10^{-3}))$; the second start is $(1, \kappa_0)$. The 10^{-3} value is an initialization floor only. For M4 and M5, $\lambda_\theta = 10^{-2}$ is fixed by hand and added to the total training negative log-likelihood; held-out scores are unpenalized. RBF centres are $\{0.1, 0.3, 0.5, 0.7, 0.9\}^2$ with width 0.12.

3.3 Estimation and decision logic

All parameters are estimated from the training replicates by likelihood or penalized likelihood, with σ^2 profiled analytically where possible. M1 and M2 are diagonal in the discrete sine-transform basis, so the likelihood is a sum over modal variances. M4-κ and M4-K minimize $-\ell(\theta, \kappa, \sigma^2) + \lambda_\theta \lVert\theta\rVert_2^2$ with σ^2 profiled out and gradients by automatic differentiation. M3 alternates between updating the whitened directions and strengths for a fixed stationary base and updating (σ^2, κ) for a fixed correction; the two are alternated rather than fitted sequentially because the range parameter can absorb part of the same low-frequency variance that would otherwise appear as a source component. M5 uses exact block coordinate ascent: for fixed (κ, θ) the whitened spike block is updated by the PPCA closed form with a free residual variance, jointly updating σ^2, the spike directions, and their strengths, and for a fixed spike block (κ, θ) are updated by penalized likelihood, so each accepted update is non-decreasing in the penalized objective. M5 is initialized from a flat conductivity field rather than from the M4-K solution, so that it does not inherit operator structure that M4-K may have used to explain source variability. The ranks of M3 and M5 are chosen by five-fold held-out log likelihood; $\widehat{r}_{\mathrm{MP}}$ is reported as a diagnostic only.

The two diagnostics have complementary roles: isolated upper spectral outliers indicate finite-dimensional source structure, and a non-vanishing deflated $\widehat{h}_N$ indicates persistent principal-part discrepancy. Together they give the structural logic

Spectral outliers	Deflated $\widehat{h}_N$	Corresponding model structure
present	$\to 0$	finite-rank correction (M3)
absent	non-vanishing	principal-operator adaptation (M4-K)
present	non-vanishing	combined adaptation and correction (M5)

This logic is evaluated, not run as an automated model-selection algorithm. A vanishing principal-part diagnostic does not by itself distinguish a stationary field from a weak lower-order perturbation; the M4-κ comparison is retained for that purpose.

4 Numerical study design

All experiments use the same reference operator and differ only in the component through which the population departs from it. We define the discretization and four synthetic populations, then the oracle refinement calculations, the finite-replicate diagnostic, and the model fits.

4.1 Domain, discretization, and reference process

The common setup is summarized in Table 4.1; its discrete covariance is (44) with $Q_W = h^{-2}I$. The assembled operators are symmetric, so $A^{-T} = A^{-1}$ numerically; the transpose is kept to make the construction explicit.

Table 4.1: Common setup.

Item	Value
Domain	$\Omega = (0,1)^2$, $n_x \times n_x$ interior nodes, $N = n_x^2$, $h = 1/(n_x+1)$
Boundary condition	homogeneous Dirichlet
Discretisation	finite differences: 5-point stencil for $-\nabla\cdot(K\nabla u)$, harmonic-mean edge conductance $2K_iK_j/(K_i+K_j)$
Reference operator	$A_0 = \kappa_0^2 I + L$, $\kappa_0 = 6$, $K_0 \equiv 1$, $\sigma^2 = 1$
White noise	$u = \sigma h^{-1} A^{-1}\xi$, $\xi \sim \mathcal{N}(0, I_N)$ (nodal variance h^{-2})
Covariance	$\Sigma = \sigma^2 h^{-2} A^{-1}(I + VCV^\top)A^{-T}$
Whitening	$z = hAu/\sigma$
Sampler	sparse LU of A (direct solve)
Mean	zero; fields are not sample-centred
Fields per truth	$n_{\text{train}} = 200$, $n_{\text{test}} = 100$
Seeds	main: $1000 + 10\,\text{id} + n_x$; robustness: $+7919s$, $s = 1,2,3$
Sensors (held-out score)	10 fixed points, noise s.d. $0.05\times$ reference RMS
Grids, model fits	$n_x \in \{16, 32\}$
Grids, diagnostics	$n_x \in \{16, 24, 32, 48, 64\}$

4.2 Population departures

The four populations are controlled representatives of source, lower-order, principal-part, and mixed departures (Table 4.2, Figure 4.1); departures are localized by the Gaussian bump $b_{c,r}(x) = \exp[-\|x-c\|^2/2r^2]$. Truth A is constructed directly in reference-whitened coordinates from three unit-norm Gaussian directions, so the population is exactly of the M3 form; the smooth nodal directions discretize admissible L^2 source profiles, not point sources. Truth B′ changes only the zeroth-order term, with a smooth leakage dip toward the lens

centre. Truth B changes the principal coefficient through

$$K(x) = \exp\left[b_{c_L,r_L}(x)\right], \tag{46}$$

which varies smoothly from 1 to e, so the remaining mismatch with the operator theorem is the square's nonsmooth boundary. Truth C composes exactly the B operator with exactly the A spike directions and strengths, without retuning either.

Table 4.2: The truths. $b_{c,r}(x) = \exp(-\|x-c\|^2/2r^2)$; $v_k = b_{c_k,r_w}/\|b_{c_k,r_w}\|_2$ (nodal norm).

Truth	Operator A	Spikes $VCV^\top$	Parameters
ref	$\kappa_0^2 I + L$	—	$\kappa_0 = 6$
A (wells)	$\kappa_0^2 I + L$	$\sum_k c_k v_k v_k^\top$	centres $(0.25, 0.30)$, $(0.70, 0.25)$, $(0.55, 0.75)$; $r_w = 0.05$; $c = (90, 60, 40)$
B′ (leakage)	$\mathrm{diag}(\kappa(x)^2) + L$	—	$\kappa(x)^2 = \kappa_0^2 + \gamma\, b_{c_L,r_L}$, $\gamma = -30$ (κ^2 from 36 to 6 at the centre)
B (lens)	$\kappa_0^2 I + L_K$	—	$K = \exp(\beta\, b_{c_L,r_L})$, $\beta = 1$: smooth, $K \in [1, e]$
C (wells+lens)	$\kappa_0^2 I + L_K$	as A	as A and B

Lens and leakage: $c_L = (0.5, 0.5)$, $r_L = 0.12$; off-grid robustness lens $c_L = (0.42, 0.57)$.
Robustness of the oracle diagnostic only: disc lens $K = e\,\mathbb{1}_{\|x-c_L\|\le 0.12} + \mathbb{1}_{\text{else}}$; constant $K \equiv 1.5$.

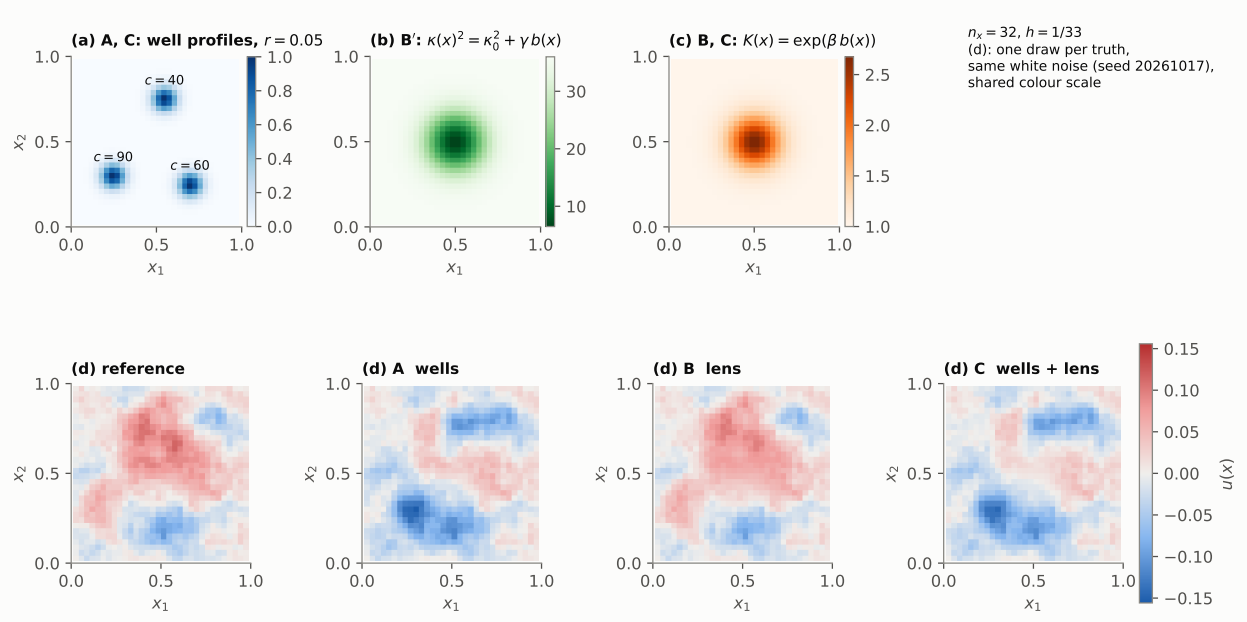


Figure 4.1: Synthetic population construction at $n_x = 32$. The top row shows the three normalized well profiles used in A and C, the spatially varying leakage coefficient used in B′, and the smooth conductivity field used in B and C. The bottom row shows one draw from the reference, A, B, and C generated with the same white-noise realization and displayed on a common colour scale.

4.3 Replicates and resolution regimes

Each main fit uses $n_{\text{train}} = 200$ independent zero-mean training fields and $n_{\text{test}} = 100$ test fields. Model fitting uses $n_x \in \{16, 32\}$ and the refinement diagnostic $n_x \in \{16, 24, 32, 48, 64\}$, so at fixed ensemble size the aspect ratio $\gamma = N/n_{\text{train}}$ grows along the refinement sequence (Table 4.3) and the diagnostic is tested in an increasingly high-dimensional regime. The well width is below one mesh spacing at $n_x = 16$, so refinement also changes the spatial resolution of the source profiles: spike strengths at $n_x = 16$ are not comparable discretizations of the finer-grid amplitudes, and the decay of source-only statistics starts from an under-resolved representation. The sample-based refinement experiment uses five independently generated training ensembles of 200 fields per truth and resolution; additional seeds are reserved for robustness checks.

Table 4.3: Regimes per grid ($n_{\text{train}} = 200$). $\sqrt{\gamma}$ is the BBP detection threshold for a whitened spike; features are resolved by r/h cells.

n_x	N	$\gamma = N/n_{\text{train}}$	$\sqrt{\gamma}$	MP edge $(1 + \sqrt{\gamma})^2$	r_w/h	r_L/h
16	256	1.28	1.131	4.543	0.85	2.04
24	576	2.88	1.697	7.274	1.25	3.00
32	1024	5.12	2.263	10.645	1.65	3.96
48	2304	11.52	3.394	19.308	2.45	5.88
64	4096	20.48	4.525	30.531	3.25	7.80

4.4 Oracle refinement quantities and predictions

Because the population covariance is known, the first experiment evaluates T_N directly, without sampling error, and records the relative spectrum, $D_N = \|T_N\|_F^2$, D_N/N, $\mathrm{KL}_N(\mu_1\|\mu_0)$, and the diagonal energy

$$h_N^{\text{oracle}} = \frac{1}{N}\sum_{i=1}^{N} T_{ii}^2, \tag{47}$$

which is the population target of $\widehat{h}_N$. For the smooth lens, midpoint quadrature of (39) from 256^2 through 2048^2 points agrees with $h_\infty = 0.060732$ at the reported precision. The finite-grid oracle values approach h_∞ from above and remain about 2.5% above it at $n_x = 64$ (Table 5.1), so $\widehat{h}_N$ is compared with h_N^{oracle} at each grid and h_∞ serves only as the asymptotic target. These quantities depend only on the configured truth and reference and were computed before any sample-based fitting.

4.5 Sample diagnostic protocol

At each truth and resolution, M1 is fitted to the 200 training fields and used as the plug-in whitening reference. Positive isolated directions are counted with the unmodified edge (27) at $\gamma = N/n_{\text{train}}$, i.e. multiplicative edge factor 1.0 with no finite-sample safety margin, so an isolated eigenvalue can occasionally cross it under the reference and the count is a diagnostic rather than a level-controlled test. The leverage correction (43) uses a denominator floor of 10^{-3}, and the variance term in (34) uses the actual number of fields in the evaluated ensemble. $\widehat{h}_N$ is reported with and without positive-spike deflation and compared with (47). Because the reference and deflation directions are estimated from the data, a cross-fitted analysis estimates them on one subset of fields and evaluates $\widehat{h}_N$ on an independent subset.

4.6 Model-fitting protocol

All models in Table 3.1 are fitted to the same training ensembles. M1 is initialized at the reference values and optimized without bounds on κ. M4-κ, M4-K and M5 share the 25 radial basis functions and the penalty weight $\lambda_\theta = 0.01$ listed under Table 3.1; the weight is fixed by hand rather than by cross-validation, and because the likelihood scales with both the number of fields and the field dimension it is small relative to the data term. The M3 PCA step fixes the whitened baseline noise variance at one, whereas the M5 PPCA step leaves it free; the M5 cycle stops on a per-cycle objective tolerance.

4.7 Evaluation metrics

Models are scored on the same 100 held-out fields per truth. The primary score is the full-field Gaussian log likelihood relative to the generating model on the same fields,

$$\Delta\ell = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left[\log p_{\widehat{\Sigma}}(u_i) - \log p_{\Sigma_1}(u_i)\right] \qquad \text{(nats per field; 0 = truth).} \tag{48}$$

The secondary score conditions on ten noisy point observations,

$$y = Hu + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_y^2 I), \tag{49}$$

with σ_y equal to 5% of the reference-field RMS, and records the conditional log predictive density of the full field, $\log p(u_{\text{test}} \mid y)$, and the RMSE of the Gaussian posterior mean; it tests whether a learned prior stays useful under sparse observations. Mechanism recovery is assessed separately: $\text{corr}(\log \widehat{K}, \log K_{\text{true}})$ and $\widehat{\kappa}$ on B and C, and the selected rank, spike strengths, per-direction cosines, and principal-angle cosines to the true source subspace on A and C.

4.8 Robustness analyses and chronology

The robustness analyses of Section 5.5 repeat the model comparison on three fresh seeds and with the lens moved off the RBF grid, repeat the oracle calculation for a disc lens and for constant $K \equiv 1.5$, vary λ_θ and the B′ replicate count, cross-fit the sample diagnostic, and compare the exact M5 fit used throughout with an earlier legacy update.

The sequence of analyses is recorded explicitly. The physical departures and the oracle refinement predictions were specified before the sample-based statistic $\widehat{h}_N$ and M5 were developed. Those two, and the explicit wrong-operator check of M4-K on the source-only truth A, were introduced afterward to address the finite-ensemble classification problem and the mixed source/operator population, and are evaluated separately from the oracle calculations.

5 Results: refinement behavior and mechanism attribution

The results address four questions: whether refinement recovers the continuum distinction when the population covariance is known exactly, whether the distinction survives estimation from 200 replicated fields, what happens when a departure is forced into the wrong structural component, and whether the combined model uses its two components selectively. The emphasis is on mechanism attribution rather than on predictive ranking alone.

5.1 Oracle refinement separates finite-rank, lower-order, and principal-part departures

Table 5.1 compares the exact relative covariance under refinement. Truth A behaves as a finite-rank perturbation of the reference: D_N and KL_N are independent of resolution, so D_N/N and h_N fall as N grows. The coarse-grid values are not an asymptotic rate, because the source profiles are under-resolved at $n_x = 16$; the structural observation is that the total relative covariance energy is bounded. The leakage perturbation B′ has the same signature without being finite rank: D_N and KL_N approach finite limits while D_N/N and h_N tend to zero, as expected of an infinite-rank but Hilbert–Schmidt departure.

The conductivity lens B is extensive: D_N/N is nearly constant over the five grids, and a least-squares line gives $\mathrm{KL}_N \approx 0.16 + 0.0315N$ with log–log slope 1.001. Table 5.1 also checks the two quantitative predictions of Lemma 2.2. First, the gap between D_N/N and h_N shrinks from 8.3×10^{-4} to 6×10^{-5} over the five grids (from 1.3% to 0.1% of h_N), so the off-diagonal relative covariance energy per node vanishes, as it must for multiplication plus a compact remainder, and h_N approaches h_∞ from above. Second, the lemma predicts the KL slope

$$\frac{1}{2|\mathcal{D}|}\int_{\mathcal{D}} \big[m - \log(1+m)\big]\, dx = 0.030796 \tag{50}$$

for the lens; the exact KL_N/N lie 2–3% above it, the same relative offset as h_N above h_∞ at $n_x = 64$, so the fitted slope is a prediction rather than a fit. The smooth conductivity perturbation therefore realizes the persistent-mode behavior of Section 2.5 rather than merely violating equivalence abstractly.

The L^2-Wasserstein distance gives a contrast: for B it stays near 4×10^{-3} while KL_N grows linearly, so the fields remain close in an ordinary transport metric while their covariance ratio accumulates an extensive high-frequency discrepancy. Truth C superposes the two mechanisms: at the available resolutions D_N/N is dominated by the finite-rank source component and decreases, while KL_N grows and h_N approaches the lens contribution from above, so the different summaries separate the finite-dimensional and distributed parts of the mixed departure.

5.2 The refinement distinction survives finite sampling

The sample experiment replaces the population covariance by 200 replicated fields whitened with a fitted stationary M1 (Figure 5.1, Table 5.2). The two-scale diagnostic retains the oracle distinction despite both finite-sample covariance noise and reference fitting.

Under the reference population, $\widehat{h}_N$ fluctuates around zero at all resolutions; the unmodified Marchenko–Pastur edge occasionally admits one apparent upper outlier, consistent with the role of $\widehat{r}_{\mathrm{MP}}$ as a diagnostic rather than a level-controlled test (Section 4.5). On A, exactly three upper outliers are detected in every replicate at every resolution; the raw statistic is dominated by the localized source directions, and after deflation $\widehat{h}_N$ tends toward zero. This holds even though the stationary whitening fit drives $\widehat{\kappa}$ toward the boundary: a plug-in reference badly distorted in its low-frequency range parameter does not produce a persistent fine-scale diagonal plateau. The weak leakage case B′ gives no stable outlier count, and both raw and deflated $\widehat{h}_N$ fluctuate around zero at the finer grids.

B instead remains nearly resolution invariant and close to the finite-grid oracle targets. For the stiff lens $K \geq 1$ the leading principal-part contribution to the lens block of T is non-positive, so the positive directions deflated on B are occasional finite-sample upper-edge crossings rather than lens signal; after removing them the estimate is within 1% of the independently specified h_∞. The persistent principal-part signature is therefore visible from replicated fields without access to Σ_1.

The mixed truth C is the more stringent test, because the fitted stationary reference again collapses its range and the raw statistic is dominated by three source directions. These are recovered as upper outliers in every replicate, and after their removal $\widehat{h}_N$ approaches the lens-scale plateau seen under B rather than tending to zero as under A: deflation exposes the distributed principal-part component beneath the finite-dimensional spikes.

Table 5.1: Oracle refinement. $D_N = \|T\|^2_{\mathrm{HS}}$, $\mathrm{KL}_N = \mathrm{KL}(\mu_1\|\mu_0)$, $h_N = \frac{1}{N}\sum_i T_{ii}^2$, W_2 in $L^2(\Omega)$ units. Lens limit $h_\infty = 0.060732$.

Truth	n_x	N	D_N	D_N/N	KL_N	h_N	W_2
A	16	256	13300.00	51.9531	88.83	11.83470	0.0754
	24	576	13300.00	23.0903	88.83	2.35010	0.0748
	32	1024	13300.00	12.9883	88.83	0.75929	0.0746
	48	2304	13300.00	5.7726	88.83	0.15306	0.0745
	64	4096	13300.00	3.2471	88.83	0.04893	0.0744
B′	16	256	0.61	0.0024	0.11	0.00058	0.0040
	24	576	0.60	0.0010	0.11	0.00015	0.0040
	32	1024	0.59	0.0006	0.11	0.00006	0.0040
	48	2304	0.59	0.0003	0.11	0.00001	0.0040
	64	4096	0.59	0.0001	0.11	4.97×10^{-6}	0.0040
B	16	256	16.54	0.0646	8.03	0.06380	0.0043
	24	576	36.88	0.0640	18.30	0.06363	0.0042
	32	1024	65.01	0.0635	32.54	0.06325	0.0042
	48	2304	144.61	0.0628	72.88	0.06266	0.0041
	64	4096	255.33	0.0623	129.01	0.06228	0.0041
C	16	256	11099.86	43.3588	87.61	10.22121	0.0707
	24	576	11149.76	19.3572	98.00	2.02637	0.0701
	32	1024	11193.69	10.9313	112.30	0.68665	0.0699
	48	2304	11285.31	4.8981	152.69	0.18169	0.0697
	64	4096	11400.42	2.7833	208.84	0.09730	0.0697

W_2 at $n_x = 24, 48$ computed for this table with the same routine as the stored values (`run_robust_T.w2`; it reproduces the stored $n_x = 16$ values to 10^{-15}). The reference is identically zero at every grid and is omitted.

Because the deflation directions are estimated from the same fields, the absolute values are a bias-corrected diagnostic rather than an exactly unbiased estimate; the refinement pattern is the principal result.

5.3 Wrong-class models fit by reallocating variability to the wrong mechanism

When a model class cannot represent the true mechanism directly, the question is which of its available components absorbs the discrepancy. Figure 5.2 and Tables 5.3–5.5 show that the resulting failures are systematic rather than simple losses in predictive score.

On the source-only truth A, the stationary model absorbs the low-frequency effect of the wells by driving its range parameter to the boundary, with $\widehat{\kappa}$ of order 10^{-5} against the generating $\kappa_0 = 6$ (Table 5.4). M2 improves held-out likelihood with a flexible stationary spectrum but follows the same collapsed range. M4-K misattributes differently, placing conductivity structure around the wells while again collapsing $\widehat{\kappa}$ (Figure 5.2a, Table 5.3). Joint source correction resolves both failures: M3-joint selects rank three and returns $\widehat{\kappa}$ close to 6. M3-nested, which fits the stationary reference first and adds spikes afterward, inherits the collapsed range, underestimates all three spike strengths, and recovers the source subspace less well; the joint fit instead overshoots the strengths (Table 5.5), as expected from upward sample-eigenvalue inflation in spiked covariance models (Baik and Silverstein, 2006). A structurally appropriate correction can thus still be distorted if the reference is estimated first and allowed to absorb the same variation.

The converse failure appears under the principal-part truth B. M1, M2 and M3-joint cannot represent the

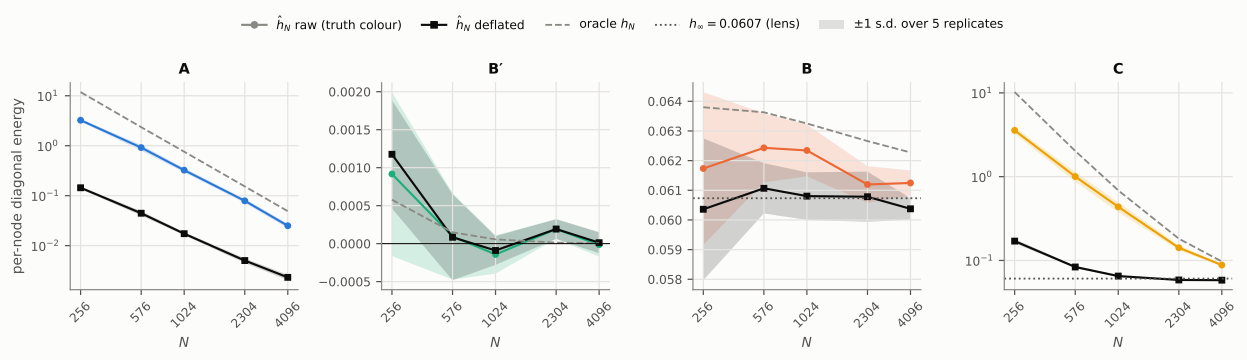


Figure 5.1: Diagonal refinement statistic $\widehat{h}_N$ from 200 fields whitened by a fitted M1, raw (truth colour) and after deflating the eigenvalues above the Marchenko–Pastur edge (black); bands show ±1 standard deviation over five replicates. Dashed curves show the oracle h_N. Dotted curves for B and C show $h_\infty = 0.0607$. On A and B′ the deflated statistic decreases under refinement; on B and C it remains at the lens scale.

Table 5.2: Diagonal refinement statistic from 200 training fields, whitened by a fitted M1 (5 replicates, mean ± s.d.). r_{MP}: eigenvalues above $(1+\sqrt{N/200})^2$ (edge factor 1.0); $\hat{h}_N = \hat{H}/N$ before and after deflating those r_{MP} directions.

Truth	n_x	r_{MP}	$\hat{\kappa}$	$\hat{\sigma}^2$	$\hat{h}_N$ raw	$\hat{h}_N$ deflated
ref	16	0.0 ± 0.0	6.04 ± 0.19	1.003 ± 0.006	0.0002 ± 0.0004	0.0002 ± 0.0004
	24	0.0 ± 0.0	5.98 ± 0.15	1.000 ± 0.005	-0.0003 ± 0.0004	-0.0003 ± 0.0004
	32	0.2 ± 0.4	6.01 ± 0.08	1.000 ± 0.002	0.0001 ± 0.0004	0.0001 ± 0.0004
	48	0.2 ± 0.4	6.10 ± 0.10	1.000 ± 0.002	0.0001 ± 0.0004	0.0002 ± 0.0004
	64	0.0 ± 0.0	6.00 ± 0.07	1.001 ± 0.001	-2.93×10^{-5} ± 0.0002	-2.93×10^{-5} ± 0.0002
A	16	3.0 ± 0.0	0.00163 ± 0.00193	1.437 ± 0.032	3.2405 ± 0.2088	0.1440 ± 0.0092
	24	3.0 ± 0.0	$(1.52 \pm 1.08) \times 10^{-4}$	1.186 ± 0.017	0.9177 ± 0.1201	0.0445 ± 0.0046
	32	3.0 ± 0.0	$(8.34 \pm 6.30) \times 10^{-5}$	1.098 ± 0.007	0.3239 ± 0.0322	0.0173 ± 0.0013
	48	3.0 ± 0.0	$(1.33 \pm 0.83) \times 10^{-4}$	1.041 ± 0.003	0.0789 ± 0.0077	0.0050 ± 0.0005
	64	3.0 ± 0.0	$(2.01 \pm 0.69) \times 10^{-4}$	1.023 ± 0.002	0.0250 ± 0.0028	0.0023 ± 0.0003
B′	16	0.6 ± 0.8	5.45 ± 0.16	0.993 ± 0.009	0.0009 ± 0.0011	0.0012 ± 0.0007
	24	0.2 ± 0.4	5.47 ± 0.12	1.000 ± 0.003	9.65×10^{-5} ± 0.0006	8.36×10^{-5} ± 0.0006
	32	0.2 ± 0.4	5.49 ± 0.15	0.997 ± 0.005	-0.0001 ± 0.0003	-9.02×10^{-5} ± 0.0002
	48	0.0 ± 0.0	5.59 ± 0.11	0.999 ± 0.002	0.0002 ± 0.0001	0.0002 ± 0.0001
	64	0.2 ± 0.4	5.52 ± 0.08	0.999 ± 0.001	-1.36×10^{-5} ± 0.0002	1.27×10^{-5} ± 0.0001
B	16	0.6 ± 0.5	5.96 ± 0.15	0.864 ± 0.005	0.0617 ± 0.0026	0.0604 ± 0.0024
	24	0.8 ± 0.4	6.04 ± 0.17	0.874 ± 0.003	0.0624 ± 0.0012	0.0611 ± 0.0008
	32	1.2 ± 0.4	5.94 ± 0.09	0.871 ± 0.003	0.0623 ± 0.0009	0.0608 ± 0.0008
	48	0.4 ± 0.5	6.01 ± 0.12	0.875 ± 0.003	0.0612 ± 0.0006	0.0608 ± 0.0008
	64	1.0 ± 0.6	6.06 ± 0.09	0.876 ± 0.001	0.0612 ± 0.0004	0.0604 ± 0.0004
C	16	3.0 ± 0.0	$(0.741 \pm 1.247) \times 10^{-3}$	1.254 ± 0.029	3.5717 ± 0.3379	0.1702 ± 0.0100
	24	3.0 ± 0.0	0.00442 ± 0.00359	1.034 ± 0.013	1.0032 ± 0.1295	0.0835 ± 0.0026
	32	3.0 ± 0.0	$(8.16 \pm 5.11) \times 10^{-5}$	0.964 ± 0.010	0.4367 ± 0.0596	0.0651 ± 0.0017
	48	3.0 ± 0.0	0.00331 ± 0.00401	0.915 ± 0.003	0.1418 ± 0.0097	0.0585 ± 0.0004
	64	3.0 ± 0.0	0.00158 ± 0.00291	0.898 ± 0.002	0.0881 ± 0.0030	0.0582 ± 0.0001

Entries are mean ± s.d. over five replicate ensembles. On A and C the M1 $\hat{\kappa}$ collapses to 10^{-5}–10^{-2}; scientific notation is used where needed to retain the spread.

conductivity field (CV correctly selects $r = 0$ for M3), and their held-out deficit grows by a factor of 4.14 when N grows fourfold, matching the extensive oracle KL; M4-K stays near the truth and recovers $\log K$ with correlation close to one. M4-κ improves the score only modestly while its coefficient ratio $\kappa(x)^2/\widehat{\kappa}^2$ spans nearly three orders of magnitude at $n_x = 32$; the audit in the footnote of Table 5.4 confirms a converged fit, so this is a lower-order model using extreme coefficient variation to imitate a principal-part change.

The weak leakage truth B′ is the complementary control. M4-κ is the structurally matched model and the only one with a score indistinguishable from the truth at both resolutions, but all B′ differences are small relative to their paired standard errors, so this deliberately weak case supports the structural classification through refinement more than through predictive ranking.

On the mixed truth C, forcing both mechanisms into either component alone fails. M3-joint captures the three wells but leaves the lens unmodeled. M4-K instead absorbs the source variation into the operator: $\widehat{\kappa}$ collapses and the conductivity-shape correlation stays well below that of M5, despite a peak $\widehat{K}$ close to the true amplitude. Predictive adequacy alone would obscure the structural source of the error; the fitted parameters show which mechanism each wrong-class model uses to compensate.

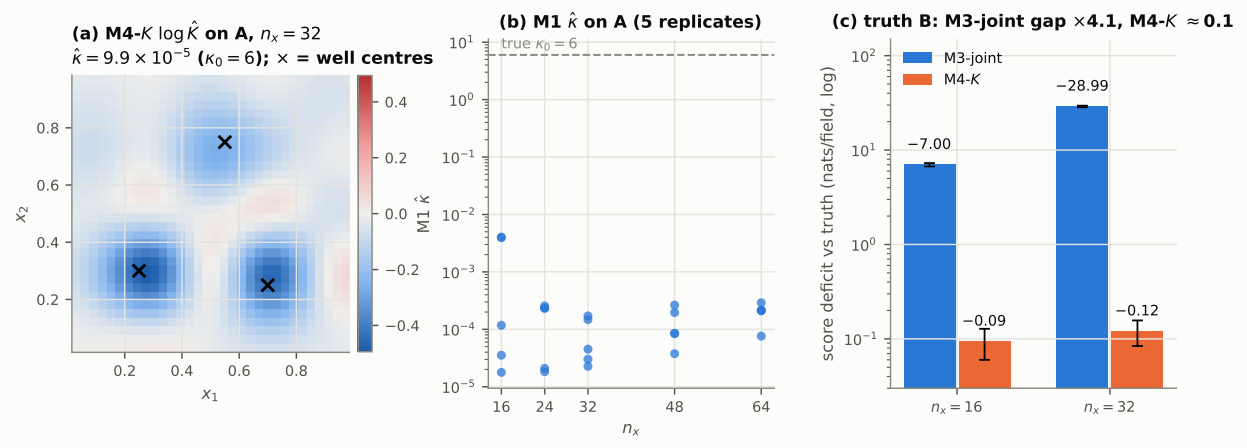


Figure 5.2: Misattribution by wrong-class models. (a) M4-K fitted to truth A at $n_x = 32$ places conductivity structure around the wells and collapses the range ($\widehat{\kappa} = 9.9 \times 10^{-5}$). (b) M1 drives $\widehat{\kappa}$ toward zero on A at every grid. (c) On truth B, the held-out deficit of M3-joint grows from 7.00 to 28.99 nats per field as N grows fourfold, while M4-K remains near -0.1. Error bars are paired standard errors.

5.4 The combined model separates source and operator variability

M5 is intended to prevent the compensating failures above by learning the principal operator and the finite-rank correction together. Figure 5.3 shows that the two components are used selectively rather than both being activated indiscriminately. In the diagnostic control at $n_x = 32$, deflation removed about one upper-edge direction on B, where these are sampling fluctuations, and exactly three on C, where they are the source directions (Table 5.2).

On A, CV selects $r = 3$, the fitted conductivity remains nearly flat, and the range returns to $\widehat{\kappa} = 6.08$ at $n_x = 32$; on the same data M1 alone gives 3.0×10^{-5}, so allowing the source component moves the inferred range by about five orders of magnitude back to $\kappa_0 = 6$. The held-out score and the recovered source subspace are essentially those of M3-joint (Tables 5.3 and 5.5), so the extra operator component does not create spurious conductivity structure when the source correction already explains the variation. On B, CV selects $r = 0$, the finite-rank component switches off, and the fit coincides with M4-K, as expected when no source correction is required. On C both components are needed: M5 selects $r = 3$, recovers the conductivity

Table 5.3: Held-out log-likelihood relative to the truth (nats per field, 100 test fields; 0 = truth) at $n_x = 16$ and 32. Superscript: CV-selected rank. Bold: best model. ×: gap growth, score at 32 divided by score at 16 (N grows 4×).

	A			B′			B			C		
Model	16	32	×	16	32	×	16	32	×	16	32	×
M1	-53.10	-60.80	1.15	-0.12	-0.14	1.11	-7.00	-28.99	4.14	-60.32	-81.96	1.36
M2	-39.14	-40.84	1.04	-0.11	-0.14	1.28	-6.99	-28.98	4.14	-47.06	-66.39	1.41
M3-joint	**-1.76**$^{(3)}$	**-5.78**$^{(3)}$	3.27	-0.12$^{(0)}$	-0.14$^{(0)}$	1.11	-7.00$^{(0)}$	-28.99$^{(0)}$	4.14	-8.44$^{(3)}$	-34.97$^{(3)}$	4.15
M3-nested	-13.34$^{(3)}$	-10.68$^{(3)}$	0.80	-0.12$^{(0)}$	-0.14$^{(0)}$	1.11	-7.00$^{(0)}$	-28.99$^{(0)}$	4.14	-22.58$^{(3)}$	-39.52$^{(3)}$	1.75
M4-κ	-46.56	-60.80	1.31	**0.01**	**0.00**	0.09	-5.90	-26.54	4.50	-54.84	-78.50	1.43
M4-K	-31.11	-47.42	1.52	-0.11	-0.25	2.30	-0.09	**-0.12**	1.28	-30.40	-42.24	1.39
M5	-1.83$^{(3)}$	-5.86$^{(3)}$	3.20	-0.11$^{(0)}$	-0.25$^{(0)}$	2.30	**-0.09**$^{(0)}$	-0.12$^{(0)}$	1.28	**-1.83**$^{(3)}$	**-6.37**$^{(3)}$	3.49

Paired standard errors (CSV) range from 0.03 to 5.6 nats; largest for the misspecified models on A and C. On B′, all models are within ≤ 0.06 nats standard error, so the apparent 0.01/0.00 advantage of M4-κ is not treated as a significant ranking and the growth ratio is not meaningful.

Table 5.4: Operator recovery. corr = corr($\log \hat{K}, \log K_{\text{true}}$) and peak $\hat{K}$ (true peak $e = 2.718$) for M4-K and M5 on B and C. Range: $\hat{K}$ for M4-K, M5; $\kappa(x)^2/\hat{\kappa}^2$ for M4-κ.

		$n_x = 16$					$n_x = 32$				
Truth	Model	$\hat{\kappa}$	corr	peak $\hat{K}$	range	r_{cv}	$\hat{\kappa}$	corr	peak $\hat{K}$	range	r_{cv}
A	M1	7.17×10^{-5}	—	—	—	—	3.05×10^{-5}	—	—	—	—
	M3-joint	6.18	—	—	—	3	6.21	—	—	—	3
	M4-κ	18.47	—	—	0.000337–0.668	—	4.45×10^{-5}	—	—	0.999–1	—
	M4-K	5.44×10^{-5}	—	—	0.31–1.01	—	9.93×10^{-5}	—	—	0.61–1.06	—
	M5	6.29	—	—	0.999–1.07	3	6.08	—	—	0.943–0.983	3
B′	M1	5.54	—	—	—	—	5.44	—	—	—	—
	M3-joint	5.54	—	—	—	0	5.44	—	—	—	0
	M4-κ	6.20	—	—	0.243–1.36	—	3.24	—	—	0.478–7.12	—
	M4-K	5.66	—	—	0.994–1.1	—	5.51	—	—	1–1.06	—
	M5	5.66	—	—	0.994–1.1	0	5.51	—	—	1–1.06	0
B	M1	6.02	—	—	—	—	6.05	—	—	—	—
	M3-joint	6.02	—	—	—	0	6.05	—	—	—	0
	M4-κ	1.07	—	—	0.0121–126	—	0.48	—	—	1.31–857	—
	M4-K	5.93	0.997	2.46	0.923–2.46	—	5.99	0.999	2.67	0.981–2.67	—
	M5	5.93	0.997	2.46	0.923–2.46	0	5.99	0.999	2.67	0.981–2.67	0
C	M1	3.07×10^{-5}	—	—	—	—	1.77×10^{-4}	—	—	—	—
	M3-joint	5.84	—	—	—	3	6.18	—	—	—	3
	M4-κ	31.88	—	—	0.000195–0.242	—	5.63	—	—	2.04e-09–2.94	—
	M4-K	3.66×10^{-5}	0.613	2.43	0.312–2.43	—	1.11×10^{-4}	0.886	2.75	0.652–2.75	—
	M5	5.77	0.997	2.52	0.956–2.52	3	6.06	0.999	2.61	0.953–2.61	3

corr is left blank on A and B′: their true conductivity is $K \equiv 1$, so $\log K_{\text{true}}$ is constant and the correlation is undefined. M4-κ on B: the extreme range $\kappa(x)^2/\hat{\kappa}^2$ is a converged fit, not an optimiser failure (433 and 597 iterations; final gradient 1.5×10^{-5} and 4.3×10^{-5} per field; restarting L-BFGS from the stored point lowers the objective by $< 10^{-10}$ per field; `data/check_m4kappa_B.json`). It is the zeroth-order model's best imitation of a principal-part change (single start).

Table 5.5: Spike recovery on A and C (true $c = 90/60/40$). $\hat{c}_k = \hat{s}_k - 1$; per-well cosines and principal-angle cosines to the true 3-d well span.

Truth	Model	n_x	r_{MP}	$\hat{c}_{1..3}$	$\cos_{1..3}$	span cosines
A	M3-joint	16	3	99.2/62.5/41.6	0.945/0.945/0.979	0.994/0.990/0.985
		32	4	101.8/77.3/47.6	0.955/0.946/0.938	0.973/0.964/0.941
	M3-nested	16	3	46.4/28.8/19.1	0.924/0.920/0.952	0.977/0.970/0.953
		32	3	63.1/47.2/29.9	0.918/0.905/0.895	0.946/0.931/0.896
	M5	16	—	98.8/61.4/41.5	0.947/0.947/0.979	0.994/0.989/0.985
		32	—	102.8/75.9/47.7	0.958/0.948/0.938	0.973/0.963/0.941
C	M3-joint	16	4	116.4/68.8/41.9	0.987/0.984/0.977	0.992/0.989/0.978
		32	4	112.6/65.6/46.0	0.875/0.859/0.934	0.974/0.955/0.936
	M3-nested	16	3	53.1/31.2/18.8	0.964/0.958/0.951	0.974/0.964/0.951
		32	3	69.1/41.7/28.6	0.853/0.834/0.894	0.941/0.922/0.894
	M5	16	—	109.7/66.0/44.1	0.987/0.984/0.984	0.994/0.990/0.985
		32	—	104.5/62.0/48.5	0.875/0.858/0.945	0.975/0.954/0.946

M3 cosines are from the rank-3 fit (the CV choice is 3 in every case shown). M5 has no r_{MP} (rank by CV only).

shape and the reference range, and retains a three-dimensional source subspace, with a held-out deficit several times smaller than those of M3-joint and M4-K. The gain is not simply additional flexibility: the decomposition simultaneously restores the reference range, the conductivity shape, and the finite-dimensional source subspace that each single-component model confounds.

M5 is not intended to replace the lower-order model. On B′, CV selects $r = 0$ and the conductivity field stays close to flat, but M5 contains no spatially varying leakage coefficient and M4-κ remains the matched model; M5 is a source-plus-principal architecture rather than a universal covariance approximator. The exact block-coordinate implementation leaves the conclusions unchanged relative to the earlier legacy alternation (Table D.1): on C at $n_x = 32$ the held-out score changes only from -6.377 to -6.373 nats per field and the conductivity correlation remains 0.9994, while $\widehat{\kappa}$ moves from 5.95 to 6.06, so the correction affects the internal consistency of the decomposition more than the model comparison.

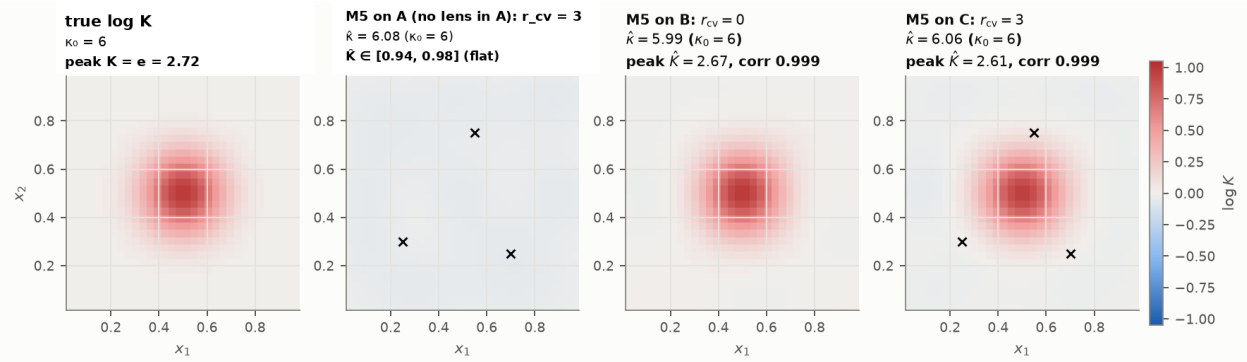


Figure 5.3: M5 selectivity at $n_x = 32$: learned $\log \widehat{K}$ and CV rank ($\kappa_0 = 6$; true peak $K = e$). The first panel shows the true $\log K$ used in truths B and C; truth A has no conductivity lens. A: $r = 3$, $\widehat{\kappa} = 6.08$, nearly flat $\widehat{K}$. B: $r = 0$, $\widehat{\kappa} = 5.99$, peak $\widehat{K} = 2.67$. C: $r = 3$, $\widehat{\kappa} = 6.06$, peak $\widehat{K} = 2.61$.

5.5 Robustness

The conclusions survive each robustness check; the tables are collected in Appendix D. Across the main seed and three fresh seeds at $n_x = 32$, M5 selects ranks 3, 0 and 3 on A, B and C (Table D.2). Moving the lens off the RBF grid increases the M5 deficits on B and C and lowers the lens correlation to about 0.974, but leaves the rank allocation unchanged. Sweeping λ_θ from 10^{-3} to 10 leaves the B and C correlations, ranks, and scores unchanged at the reported precision (Table D.5). Cross-fitting the whitening and deflation on independent 100-field halves preserves the plateau on B and C and the decay on A (Table D.3), so the separation does not depend on same-sample fitting. Increasing the B′ ensemble from 200 to 800 fields raises the M4-κ leakage correlation while all score differences stay below 0.15 nats per field (Table D.4): the weak lower-order case becomes more recoverable with replication. With ten sensors at 5% noise, M5 on C keeps a posterior-mean RMSE close to that of the truth and a conditional deficit several times smaller than M3-joint or M4-K (Table D.7), so the mechanism allocation remains useful under sparse conditioning.

The principal-part signature also persists for coefficients other than the smooth lens (Table D.6). For the discontinuous disc lens, which Lemma 2.2 does not cover, D_N/N stays non-vanishing and KL_N grows linearly. The constant rescaling $K \equiv 1.5$ gives a second, independent check of the lemma's plateau constant: here $m = K^{-2} - 1$ is constant, so the lemma predicts $D_N/N \to (1/2.25 - 1)^2 = 0.3086$, and the computed values rise from 0.2919 to 0.3070 toward it. Its KL slope agrees likewise: $\mathrm{KL}_N/N = 0.1268$ at $n_x = 64$ against the predicted $\frac{1}{2}[m - \log(1 + m)] = 0.1277$. The lemma's constants are thus checked on two principal-part departures of different form.

6 Discussion and conclusion

The central result is that replicated fields can decide whether a discrepancy from a Gaussian reference should be represented as a correction of that reference or as a change in its covariance-generating operator. Source and lower-order departures produced refinement behavior consistent with the reference measure class, a principal-coefficient change produced persistent relative covariance energy and an extensive KL cost, and $\widehat{h}_N$ turned this distinction into a sample quantity that, for the lens, tracked the oracle sequence and the predicted h_∞. Wrong-class models did not merely lose score; they reassigned variability to plausible but incorrect components, and M5, given both components, allocated each form of variability to the right one.

The resulting modeling rule is: *learn the operator where the population demands a change of Gaussian measure class; correct relative to that operator where the remaining discrepancy stays within class*. When a discrepancy persists across an increasing fraction of resolved modes, adding correction terms to a fixed reference does not address it; the reference itself must change.

Implications for learned priors and function-space generative models. Discretization-independent generative constructions already make absolute-continuity and Cameron–Martin assumptions explicit (Lim et al., 2025; Pidstrigach et al., 2023). If the principal part of the Gaussian reference is wrong, no finite-dimensional correction repairs the measure-class mismatch, even though a fixed mesh can make the corrected covariance look adequate. A refinement diagnostic of the type used here can serve as a pre-training check on the reference.

Implications for reference-based complexity priors. For a principal-part mismatch, $\mathrm{KL}_N = O(N)$, so the PC distance $d_N = \sqrt{2\,\mathrm{KL}_N}$ grows as $O(\sqrt{N})$ and the same physical departure looks more complex as the mesh is refined (Simpson et al., 2017; Fuglstad et al., 2019). A discretization-stable complexity then needs either a different distance, as for singular operator changes (Llamazares-Elias et al., 2026), or a new base operator relative to which the remaining departure is within class.

Implications for spatial covariance fitting. With unmodeled wells, the stationary fit gives $\widehat{\kappa} \approx 3 \times 10^{-5}$ against a generating value of 6, while M5 returns $\widehat{\kappa} = 6.08$. The base of the Matérn PC prior for the range is infinite range, $\kappa = 0$ (Fuglstad et al., 2019), so shrinkage toward it would reinforce the confounded direction. A near-zero fitted range when sources may be unmodeled is a mechanism-attribution warning, not automatically evidence of long-range dependence.

Limitations. Several robustness checks in Section 5.5 reduce implementation-specific concerns: although λ_θ was fixed by hand, sweeping it from 10^{-3} to 10 leaves the B and C M5 ranks and scores unchanged at the reported precision; across four seeds the $n_x = 32$ M5 ranks remain 3/0/3 for A/B/C; moving the lens off the RBF grid lowers the fitted lens correlation to about 0.974 but preserves the B/C rank allocation; a discontinuous disc lens retains non-vanishing D_N/N and a growing KL cost; and the constant rescaling $K \equiv 1.5$ likewise retains extensive principal-part scaling. The remaining limitations are structural. The source-deflation rule was evaluated with a stiff conductivity lens, $K \geq 1$, so although the diagonal refinement statistic h_N is sign insensitive, the upper-edge source-removal step is not yet a general soft-lens classifier. The study is two-dimensional and employs a square domain whose corners fall outside the smooth-boundary assumptions of the strongest variable-coefficient equivalence theorem. M5 does not include a spatially varying lower-order coefficient, and M4 fits use a single initialization. No real aquifer data are analyzed, and the sample-based diagonal refinement statistic $\widehat{h}_N$ and M5 architecture were developed after the oracle refinement results rather than specified in advance.

Next steps. A soft-lens experiment would separate the sign-insensitive diagnostic from the sign-specific deflation rule, and low-replicate regimes would show how much replication the classification needs. Lemma 2.2 should be extended to discontinuous coefficients, as the disc lens suggests, to tensor-valued K, and to $d = 3$. Samplers for principal-part hyperparameters, which index mutually singular families, deserve study under refinement. Finally, the approach should be tested on replicated observations from a pumped heterogeneous aquifer, and on population priors over constitutive behavior transferred between experiments in mechanics.

The opening question can therefore be answered empirically: whether to correct a Gaussian reference or change its operator is a question that replicated data can answer, and getting that decision wrong can produce a confident, physically plausible, but structurally incorrect model.

A Proof of Proposition 2.1

Proof. If $T \in \mathcal{S}_2$ and $P_N \to I$ strongly, then finite-rank compression converges in Hilbert–Schmidt norm,

$$\|P_N T P_N - T\|_{\mathrm{HS}} \longrightarrow 0. \tag{51}$$

Therefore

$$D_N = \|P_N T P_N\|_{\mathrm{HS}}^2 \longrightarrow \|T\|_{\mathrm{HS}}^2 < \infty. \tag{52}$$

Since the numerator converges to a finite limit while $N \to \infty$,

$$\frac{D_N}{N} \longrightarrow 0. \tag{53}$$

□

B Proof of Lemma 2.2

Proof. Write $\mathcal{K}$ for the compact operators on $L^2(\mathcal{D})$.

Step 1 (regularity). For a smooth or convex polygonal domain and $K \in C^1(\overline{\mathcal{D}})$, $K \geq k_0 > 0$, both A_0^{-1} and A_1^{-1} map $L^2(\mathcal{D})$ boundedly into $H^2(\mathcal{D}) \cap H_0^1(\mathcal{D})$. Hence $A_0 A_1^{-1}$ and $A_1 A_0^{-1}$ extend to bounded operators on $L^2(\mathcal{D})$, and each is the inverse of the other.

Step 2 ($A_1 A_0^{-1} = M_K + \mathcal{K}$). Let $B = A_1 - A_0 = -\nabla\cdot((K-1)\nabla) = -(K-1)\Delta - \nabla K \cdot \nabla$. Since $-\Delta A_0^{-1} = I - \kappa^2 A_0^{-1}$,

$$BA_0^{-1} = M_{K-1} - \kappa^2 M_{K-1} A_0^{-1} - (\nabla K \cdot \nabla) A_0^{-1}. \tag{54}$$

The operator A_0^{-1} is compact on L^2, and ∇A_0^{-1} maps L^2 boundedly into H^1, which embeds compactly into L^2 (Rellich). Multiplication by the bounded functions $K-1$ and ∇K preserves compactness. Hence $BA_0^{-1} = M_{K-1} + C_1$ with $C_1 \in \mathcal{K}$, and $A_1 A_0^{-1} = I + BA_0^{-1} = M_K + C_1$.

Step 3 ($A_0 A_1^{-1} = M_{1/K} + \mathcal{K}$). Let $G = A_0 A_1^{-1}$. From $(M_K + C_1)G = I$,

$$G = M_{1/K} - M_{1/K} C_1 G = M_{1/K} + C_2, \qquad C_2 \in \mathcal{K}. \tag{55}$$

Step 4 (the decomposition). Since $\Sigma_0^{-1/2} = c^{-1/2} A_0$ and $\Sigma_1 = c\, A_1^{-2}$, and A_0, A_1 are self-adjoint,

$$T + I = A_0 A_1^{-1} A_1^{-1} A_0 = GG^* = (M_{1/K} + C_2)(M_{1/K} + C_2^*) = M_{1/K^2} + C_3, \qquad C_3 \in \mathcal{K}, \tag{56}$$

which is (17) with $R = C_3$. Since G is invertible, $T + I \geq \gamma I$ for some $\gamma > 0$.

(i). By Weyl's theorem, compact perturbations do not change the essential spectrum, so $\sigma_{\text{ess}}(T) = \sigma_{\text{ess}}(M_m)$, the essential range of m, which equals $m(\overline{\mathcal{D}})$ for continuous m. A compact operator has essential spectrum $\{0\}$, so T is compact only if $m \equiv 0$, i.e. $K \equiv 1$.

(ii). In the orthonormal basis of normalized cell indicators, $P_N M_m P_N$ is diagonal with entries the cell averages $\bar{m}_i$, so $\frac{1}{N}\|P_N M_m P_N\|_{\text{HS}}^2 = \frac{1}{N}\sum_i \bar{m}_i^2 \to |\mathcal{D}|^{-1}\int m^2$ by uniform continuity of m. For the remainder, $P_N R P_N$ has rank at most N and singular values $s_j(P_N R P_N) \leq s_j(R)$, so

$$\frac{1}{N}\|P_N R P_N\|_{\text{HS}}^2 \leq \frac{1}{N}\sum_{j=1}^{N} s_j(R)^2 \longrightarrow 0, \tag{57}$$

because $s_j(R) \to 0$ and Cesàro means of a null sequence vanish. The cross term vanishes by Cauchy–Schwarz, giving the first limit in (18). The diagonal entries of T_N are $\bar{m}_i + (P_N R P_N)_{ii}$ with $\frac{1}{N}\sum_i (P_N R P_N)_{ii}^2 \leq \frac{1}{N}\|P_N R P_N\|_{\text{HS}}^2 \to 0$, giving the second limit. Because $P_N M_m P_N$ has no off-diagonal entries in this basis, the difference of the two quantities tends to zero.

(iii). By the Hoffman–Wielandt inequality, with eigenvalues and cell averages both sorted,

$$\frac{1}{N}\sum_{j=1}^{N} \left|t_{j,N} - \bar{m}_{(j)}\right|^2 \leq \frac{1}{N}\|P_N R P_N\|_{\text{HS}}^2 \longrightarrow 0, \tag{58}$$

so the empirical eigenvalue distribution of T_N has the same weak limit as that of the $\bar{m}_i$, namely the law of $m(X)$. The persistent-mode statement follows at every δ with $|\{|m| = \delta\}| = 0$. Finally, all $t_{j,N}$ and $\bar{m}_i$ lie in a compact interval $[\gamma' - 1, \|T\|]$ with $\gamma' = \min(\gamma, \min K^{-2}) > 0$, on which $f(t) = t - \log(1+t)$ is Lipschitz; hence $\frac{1}{N}\sum_j f(t_{j,N}) - \frac{1}{N}\sum_i f(\bar{m}_i) \to 0$, and the second sum tends to $|\mathcal{D}|^{-1}\int f(m)$. This gives (19). □

C Optimization settings

Table C.1: Optimiser settings (L-BFGS-B, SciPy; gradients by JAX autodiff).

Model	Inner solve	Outer loop	Other
M1, M2	maxiter 1000, maxfun 3000, gtol 10^{-8}, ftol 10^{-13}	—	—
M4-κ, M4-K	maxiter 2000, maxfun 6000, gtol 10^{-8}, ftol 10^{-13}	—	dense Cholesky
M3-joint	maxiter 500, gtol 10^{-10}, ftol 10^{-15}	$\Delta\ell < 10^{-9}$ per field, ≤ 1000	2 starts
M5	maxiter 500, gtol 10^{-8}, ftol 10^{-14}	$\Delta\ell < 10^{-7}n$ per cycle, ≤ 60	monotone check, slack $10^{-8}n$

κ floored at 10^{-3} at initialisation only; no bounds during optimisation.

D Robustness tables

Table D.1: M5: exact block coordinate ascent vs the legacy alternation (legacy shown for comparison only). Cycles and final gradient norm $\|\nabla_{(\log\kappa,\theta)}\|$ of the total objective ($n_{\text{train}} = 200$) for the exact fit; the legacy run did not record them.

Truth	n_x	score (leg.)	score (exact)	r_{cv}	$\hat{\kappa}$ (leg.)	$\hat{\kappa}$ (exact)	corr (leg.)	corr (exact)	cycles	$\|\nabla\|$
A	16	-1.83	-1.83	3/3	5.87	6.29	0.0923	0.0563	15	0.301
B′	16	-0.11	-0.11	0/0	5.66	5.66	-0.4548	-0.4548	2	0.002
B	16	-0.09	-0.09	0/0	5.93	5.93	0.9970	0.9970	2	0.001
C	16	-1.87	-1.83	3/3	5.32	5.77	0.9965	0.9974	14	0.259
A	32	-5.86	-5.86	3/3	6.00	6.08	0.0569	0.0511	8	0.301
B′	32	-0.25	-0.25	0/0	5.51	5.51	-0.1949	-0.1949	2	0.015
B	32	-0.12	-0.12	0/0	5.99	5.99	0.9990	0.9990	2	0.010
C	32	-6.38	-6.37	3/3	5.95	6.06	0.9994	0.9994	13	0.304

corr here is against the lens $\log K$ for every truth and is meaningful only for B and C (see Table 5.4).

Table D.2: Robustness: held-out score vs truth (nats per field) over the main seed and three fresh seeds (mean ± s.d.), and the off-grid lens (centre $(0.42, 0.57)$). M5 exact everywhere.

Truth	n_x	seeds	M1	M2	M3-joint	M4-κ	M4-K	M5	M5 r_{cv}
A	16	4	-52.45±1.81	-39.32±1.14	-1.74±0.23	-45.61±1.52	-30.40±0.99	-1.81±0.24	3
A	32	4	-60.64±2.00	-40.27±0.89	-6.97±0.86	-58.21±2.46	-47.17±1.30	-7.04±0.86	3
B′	16	4	-0.07±0.03	-0.08±0.02	-0.07±0.03	-0.05±0.04	-0.12±0.02	-0.12±0.02	0
B′	32	4	-0.12±0.03	-0.13±0.01	-0.12±0.03	-0.04±0.05	-0.16±0.06	-0.16±0.06	0
B	16	4	-6.82±0.20	-6.82±0.21	-6.82±0.20	-5.47±0.28	-0.09±0.02	-0.09±0.02	0
B	32	4	-28.47±0.68	-28.47±0.69	-28.47±0.68	-25.98±0.65	-0.10±0.02	-0.10±0.02	0
C	16	4	-57.54±2.04	-45.32±1.30	-8.36±0.16	-51.85±2.03	-29.36±0.96	-1.98±0.30	3
C	32	4	-88.01±4.32	-68.57±1.79	-35.47±0.44	-84.86±5.08	-46.04±2.66	-7.14±0.50	3
B off-grid	16	1	-6.99	-6.99	-6.99	-5.69	-0.37	-0.37	0
C off-grid	16	1	-60.06	-46.83	-8.67	-54.55	-31.29	-2.15	3
B off-grid	32	1	-28.63	-28.62	-28.63	-26.19	-1.73	-1.73	0
C off-grid	32	1	-81.62	-65.75	-34.36	-80.65	-44.67	-7.91	3

M3-nested was not refitted on the robustness seeds (`run_robust_fits.py` skips it); it is omitted here.

Table D.3: Cross-fitted diagnostic: M1 whitening and deflation directions fitted on 100 fields, $\hat{h}_N$ computed on the other 100, swapped and averaged (5 replicates; same fields as Table 5.2).

Truth	n_x	$\hat{h}_N$ raw	$\hat{h}_N$ deflated	r_{MP}	in-sample deflated
ref	16	0.0008 ± 0.0010	0.0008 ± 0.0010	0.0	0.0002
	24	-0.0004 ± 0.0003	-0.0004 ± 0.0003	0.2	-0.0003
	32	0.0003 ± 0.0005	0.0003 ± 0.0005	0.2	0.0001
	48	0.0001 ± 0.0004	0.0001 ± 0.0004	0.4	0.0002
	64	-4.26×10^{-5} ± 0.0002	-4.04×10^{-5} ± 0.0002	0.2	-2.93×10^{-5}
A	16	3.2633 ± 0.2080	0.1236 ± 0.0089	3.0	0.1440
	24	0.9234 ± 0.1263	0.0320 ± 0.0042	3.0	0.0445
	32	0.3286 ± 0.0312	0.0102 ± 0.0008	3.0	0.0173
	48	0.0787 ± 0.0079	0.0029 ± 0.0007	3.0	0.0050
	64	0.0250 ± 0.0025	0.0018 ± 0.0003	3.0	0.0023
B′	16	0.0008 ± 0.0013	0.0008 ± 0.0013	0.0	0.0012
	24	-7.24×10^{-5} ± 0.0005	-6.43×10^{-5} ± 0.0005	0.1	8.36×10^{-5}
	32	-0.0003 ± 0.0005	-0.0003 ± 0.0005	0.2	-9.02×10^{-5}
	48	0.0001 ± 0.0002	0.0001 ± 0.0002	0.2	0.0002
	64	-6.29×10^{-5} ± 0.0003	-6.04×10^{-5} ± 0.0003	0.3	1.27×10^{-5}
B	16	0.0621 ± 0.0030	0.0621 ± 0.0030	0.6	0.0604
	24	0.0627 ± 0.0012	0.0627 ± 0.0012	0.4	0.0611
	32	0.0621 ± 0.0007	0.0621 ± 0.0007	0.4	0.0608
	48	0.0610 ± 0.0006	0.0610 ± 0.0006	0.5	0.0608
	64	0.0613 ± 0.0006	0.0612 ± 0.0006	0.4	0.0604
C	16	3.5952 ± 0.3750	0.1512 ± 0.0092	3.0	0.1702
	24	1.0091 ± 0.1273	0.0742 ± 0.0020	3.0	0.0835
	32	0.4390 ± 0.0575	0.0614 ± 0.0014	3.0	0.0651
	48	0.1416 ± 0.0096	0.0596 ± 0.0003	3.0	0.0585
	64	0.0880 ± 0.0030	0.0605 ± 0.0003	3.0	0.0582

Table D.4: Truth B′ with more training fields (nested: the main 200 plus new fields; main test set). Scores in nats per field vs truth; M4-κ corr with the true $\log(\kappa(x)^2/\kappa_0^2)$.

n_x	n_{train}	M1	M1 $\hat{\kappa}$	M3-joint	r_{cv}	r_{MP}	M4-κ	corr
16	200	-0.125	5.54	-0.125	0	0	0.007	0.738
16	400	-0.133	5.64	-0.133	0	0	-0.024	0.856
16	800	-0.138	5.56	-0.138	0	0	-0.042	0.929
32	200	-0.138	5.44	-0.138	0	0	0.001	0.800
32	400	-0.144	5.49	-0.144	0	0	-0.038	0.861
32	800	-0.148	5.52	-0.148	0	0	-0.025	0.874

Table D.5: λ_θ sweep (penalty $\lambda_\theta\|\theta\|^2$ on the total negative log-likelihood). True peak $K = e = 2.718$.

Truth	n_x	λ_θ	M4-K corr	M4-K max $\hat{K}$	M4-K score	M5 corr	M5 max $\hat{K}$	M5 r_{cv}	M5 score
B	16	0.001	0.9970	2.46	-0.09	0.9970	2.46	0	-0.09
B	16	0.01	0.9970	2.46	-0.09	0.9970	2.46	0	-0.09
B	16	0.1	0.9970	2.46	-0.09	0.9970	2.46	0	-0.09
B	16	1	0.9970	2.45	-0.09	0.9970	2.45	0	-0.09
B	16	10	0.9973	2.40	-0.08	0.9973	2.40	0	-0.08
B	32	0.001	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	0.01	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	0.1	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	1	0.9990	2.67	-0.12	0.9990	2.67	0	-0.12
B	32	10	0.9990	2.65	-0.12	0.9990	2.65	0	-0.12
C	16	0.001	0.6131	2.43	-30.40	0.9974	2.52	3	-1.83
C	16	0.01	0.6131	2.43	-30.40	0.9974	2.52	3	-1.83
C	16	0.1	0.6131	2.43	-30.40	0.9974	2.52	3	-1.83
C	16	1	0.6134	2.44	-30.39	0.9975	2.51	3	-1.83
C	16	10	0.6159	2.50	-30.38	0.9977	2.44	3	-1.83
C	32	0.001	0.8864	2.75	-42.24	0.9994	2.61	3	-6.37
C	32	0.01	0.8864	2.75	-42.24	0.9994	2.61	3	-6.37
C	32	0.1	0.8864	2.75	-42.24	0.9994	2.61	3	-6.37
C	32	1	0.8864	2.75	-42.23	0.9994	2.61	3	-6.37
C	32	10	0.8868	2.74	-42.22	0.9994	2.60	3	-6.37

Table D.6: Oracle refinement for the disc lens and constant $K \equiv 1.5$ (smooth lens B in Table 5.1). For constant K, Lemma 2.2 predicts $D_N/N \to (K^{-2} - 1)^2 = 0.3086$.

Truth	n_x	D_N/N	KL_N	W_2
B_disk	16	0.0325	5.21	0.0030
	24	0.0403	15.87	—
	32	0.0374	26.73	0.0033
	48	0.0362	59.80	—
	64	0.0351	104.30	0.0032
K_const	16	0.2919	30.39	0.0093
	24	0.2999	70.81	—
	32	0.3032	127.70	0.0090
	48	0.3059	290.71	—
	64	0.3070	519.22	0.0089

Table D.7: Ten sensors, 5 % noise, $n_x = 32$: conditional log predictive density $\log p(u_{\text{test}} \mid y)$ of the full field given the sensor readings, relative to the truth (nats per field), and RMSE of the posterior mean field.

Model	A lpd	A RMSE	B′ lpd	B′ RMSE	B lpd	B RMSE	C lpd	C RMSE
truth	0.00	0.0347	0.00	0.0266	0.00	0.0239	0.00	0.0314
M1	-58.89	0.0564	-0.09	0.0266	-28.89	0.0240	-80.28	0.0494
M2	-37.07	0.0562	-0.11	0.0267	-28.89	0.0240	-62.87	0.0496
M3-joint	-5.77	0.0348	-0.09	0.0266	-28.89	0.0240	-34.89	0.0318
M3-nested	-9.87	0.0358	-0.09	0.0266	-28.89	0.0240	-38.65	0.0327
M4-κ	-58.89	0.0564	-0.02	0.0266	-26.07	0.0247	-76.15	0.0497
M4-K	-45.38	0.0563	-0.20	0.0266	-0.12	0.0239	-40.38	0.0501
M5	-5.85	0.0348	-0.20	0.0266	-0.12	0.0239	-6.29	0.0316

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